

transferable, and will be issued within seven days of the application. There are some minimum requirements for remote pilots. They have to be 18 years old, have to have passed the 10th standard exam, and have training from a DGCA approved Flying Training Organization (FTO). The requirements are not applicable to nano drone and micro drone flights.

Now, the permission for drone operations can only be obtained during the day time, and for line of sight flights only. This means that the drone has to be visible to the operator at all times. The maximum allowed altitude for flights is 400 feet. Drone flights cannot be conducted if the cloud ceiling is below 1500 feet, if the ground visibility is less than 5 km, and if it is raining or snowing. There are some restrictions on where a drone can be flown as well. These are near airports, defence establishments and over 500 meters into the sea from the coastline. The drone also has to be insured for any damage caused to third parties.

A detailed flight plan has to be filed 24 hours before the drone flight. The permissions for flights are to be obtained from the Digital Sky Platform. If there is no permission obtained, there cannot be a take off. The Digital Sky Platform will have colour coded zones according to the area of the intended drone flight. Operations in the red zones are completely prohibited while the yellow zones are controlled airspaces where permissions may be granted depending on the conditions and timing of request. If an area is marked as green, then the permission for the flight will be granted automatically. At this point of time, a single operator is not allowed to simultaneously operate more than one drone at a time.

The regulations also specify procedures for maintenance of the drones, and the equipment necessary on every drone. The complete text of the drone regulations is available at <https://digit.in/dr1>.

DRONE REGULATIONS 2.0

R Venkateswaran, Senior Vice President, IoT Solutions at Persistent Systems commented on various aspects of

the drone regulations: "It is commendable that the DGCA has demonstrated the right intent by announcing the Policy for Remotely Piloted Aircrafts (also known as drones) in India much before the widespread availability of such drones. This ensures that businesses that are planning to adopt drones will have sufficient clarity on the usage guidelines. Some aspects of the policy could be improved to ensure wider adoption and development of innovative drone-centric solutions.

Firstly, the drone categories "nano" & "micro" (maximum take-off weight

private blood bank delivering blood to a remote accident victim via drones during a medical emergency would need exemption from such processes.

Thirdly, "uncontrolled airspace" is a broad term and should also include minimum flying height requirements in public areas to prevent disturbance/annoyance to common public.

With some of these changes, the policy will definitely facilitate the proliferation of drone adoption across multiple businesses."

The new regulations will come into effect from 1 December, 2018. The full



Nano drone flights up to an altitude of 50 feet are not regulated

less than 2kg) are likely to be for recreational purposes only and quite restrictive and impractical for any commercial business model. Adopting the prevalent guidelines from other countries, the micro category should be updated to cover drones up to 25kg (or at least 7kg) to help different businesses to adopt drones as part of their strategy. Larger drones are needed for delivering vital medicines or preventing the spread of wild fires.

Secondly, the process of informing/seeking approval from the local police 24-hours prior to each drone operation is very onerous – both for the drone owner as well as the over worked police department. For example, a

impact of the regulations, and the minutiae of how the Digital Sky Platform will work, will only be understood after the rollout of the platform. Under the new policy regime, certain kinds of commercial uses of drones are not allowed. The prohibited use cases include parcel delivery, and air taxis. At this point, there are no specific regulations for drone swarms. However, the DGCA has indicated that it is already in the process of drafting a 2.0 version of the policy, and in the future these use cases may be permitted. The new regulations are also expected to go into certifications of drone platforms and software and cover beyond line of sight drone operations. 